

Shuvam Thakur

SOFTWARE ENGINEER

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 Sydney City, NSW  Available to work on request

PROFILE SUMMARY

Software Engineer with a proven track record in a global investment-banking technology team, specializing in full-stack development, distributed systems, and large-scale data engineering. Experienced in systems administration, production troubleshooting, Agile SDLC, and managing platform systems calculating counterparty risk. Pursuing a Master of AI at UTS to deepen expertise in Machine Learning models, NLP, and predictive analytics.

EDUCATION

University of Technology Sydney (UTS)

Master of Artificial Intelligence | July 2026 – Present

Delhi Technological University (DTU)

Bachelor of Technology in Computer Engineering (GPA: 8.23) | 2019 – 2023

PROFESSIONAL EXPERIENCE

Macquarie Global Services • Senior Associate (L2 Software Engineer)

April 2025 – June 2026

Engineer on the firm's strategic credit-risk platform, owning applications that calculate counterparty exposure and credit limits across the global trading book.

CORE DELIVERY & BUSINESS IMPACT

- **Technical SME:** Recognized as the subject-matter expert for platform calculations and data validations; served as primary engineering contact for credit-risk business users.
- **Production Stability:** Resolved 245 critical tickets (defects, stories, incidents) across the credit-risk platform, the exposure database, and downstream batch pipelines.
- **Division Upgrade:** Led the technical execution of a multi-month core data-model expansion, orchestrating 8 deployment bundles across 100+ database tables and dozens of stored procedures.
- **Infrastructure Modernization:** Contributed to a global Sybase-to-Postgres replication migration and led the division-wide PKI certificate refresh across 12+ services.
- **Technical Documentation:** Authored 36+ internal Confluence architecture pages and runbooks read by engineering teams.

TECHNICAL DEEP DIVES

- **Incident Resolution:** Traced a calculation pipeline bug through a 17-step state inconsistency between adjacent handlers, spanning five microservices and the database.
- **Data Warehouse Debugging:** Root-caused a missing rating defect in the scoring application where Hibernate dirty-checking was silently overwriting a final-rating column for ~500 records.
- **Batch Ingestion Fixes:** Resolved a duplicate data incident on a daily batch by tracing an upstream API failing to push to a downstream allocation table, producing an automated safeguard script.
- **Engineering Contribution:** Contributed to 6+ Spring Boot microservices, logged 720+ commits, and maintained an Angular microfrontend ecosystem comprising 12 micro-frontends.

Macquarie Global Services • Associate Engineer

November 2023 – March 2025

- **Backend Development:** Built and deployed enterprise-grade Spring Boot microservices to RESTful API standards, delivering validation refactors and batch job performance fixes.
- **Full-Stack SDE:** Developed responsive Angular dashboards and enforced 90%+ JUnit code coverage using JaCoCo gating.
- **DevOps & Security:** Owned Bamboo CI/CD workflows and managed application security (Fortify, Black Duck) and SSL/p12 certificate refresh via Cyberark.

Macquarie Global Services • Full-Stack SDE Intern

January 2023 – October 2023

- Collaborated with senior engineers to implement RESTful Spring Boot microservices for risk-calculation modules.
- Developed responsive Angular frontend dashboards, ensuring real-time risk data visualization.
- Configured CI/CD pipelines using Bamboo and actively participated in Agile ceremonies and code reviews.

PROFESSIONAL EXPERIENCE (CONTINUED)

Airtel Payments Bank • Backend Engineering Intern

June 2022 – August 2022

- Engineered and optimized Java Spring Boot backend microservices contributing to secure payment-processing.
- Developed responsive customer-facing banking portal interfaces utilizing Angular 8 templates.
- Integrated third-party and internal RESTful APIs, ensuring real-time consistency and high availability.

LIS • Data Engineering Intern

August 2021 – November 2021

- Architected and deployed automated Python web-scraping scripts to harvest raw text data from unstructured sources.
- Curated, cleaned, and pre-processed large-scale text datasets for Natural Language Processing (NLP) model training.

Rato Bangala Kitab • Social Media & Graphics Design Intern

June 2018 – January 2019

- Designed brand-aligned graphical content, infographics, and posters, managing regular website updates.
- Directed multimedia production and post-production cinematography editing for the RBK YouTube recital channel.

Sofos Softwares Pvt. Ltd. • Web Designer Intern

December 2017 – January 2018

- Assisted in web design and frontend development utilizing HTML, CSS, and JavaScript for interactive client pages.
- Collaborated with core developers to troubleshoot UI/UX issues, ensuring cross-browser compatibility.

SELECTED PROJECTS

Early Detection of Alzheimer's Disease via Speech Analysis [Deep Learning / NLP](#)

- **Architecture Design:** Engineered a hybrid CNN + BiLSTM (2 layers) network with max-pooling to extract linguistic features from speech transcripts.
- **Benchmark Analysis:** Achieved 92% accuracy and F1-macro of 0.92 benchmarking custom models against pre-trained transformer blocks (AlBERT) for early-stage AD in Hindi.

Multimodal Transformer Adaptation [Computer Vision / Social Computing](#)

- **Framework Customization:** Designed task-specific multimodal classifiers utilizing CLIP-ViT backbones, developing a Co-Attention Ensemble model for Hate Speech detection (F1 score: 0.7929).

TECHNICAL SKILLS

Languages & Frameworks: Java (Spring Boot, JDBC, Hibernate), Python, TypeScript, Angular, HTML/CSS, SQL (T-SQL, PL/SQL), Bash, Scala (Spark)

Systems, Cloud & DevOps: Systems Administration (Linux/Windows Server), Azure, AWS (RDS, S3, EKS, EMR, IAM), Docker, Bamboo CI/CD, Git, CyberArk, Active Directory, Server Monitoring, Fortify

Databases & Data Eng: Distributed Systems, RESTful APIs, MS SQL Server, Sybase, Oracle, Postgres, MongoDB, Apache Spark, Kafka, Airflow, Solace JMS, Presto/Trino

AI, ML & Observability: PyTorch, TensorFlow, HuggingFace, scikit-learn, NLP, Computer Vision, MLOps, Grafana, Splunk, Postman, JUnit + JaCoCo

ACHIEVEMENTS & AWARDS

- **Google Cloud Certified:** Data Science & ML Track
- **Duke of Edinburgh:** Bronze Award Recipient (2017)
- **Gold Medalist:** Mahasangram Debate Competition (2017)
- **Global Rank 6:** International Mars Hackathon (2020)
- **Second Runner-Up:** Indian Rover Challenge (2019)

REFERENCES

Karan Juneja (Manager and Credit Risk Tech Lead, Macquarie Group) • +61 412 036 110